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Algebra 2 - Tutorial 1

Basic Training Cycle

Matrices

Exercise 1

Let $A = \begin{pmatrix} 3 & 1 & -2 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{pmatrix}$.

I) Define $P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$.

- 1 Prove that P is invertible and find its inverse.
- 2 Compute $D = P^{-1}AP$, D^n then A^n .
- 3 Show that D is invertible and deduce that A is invertible.
- 4 Deduce the expression of A^{-n} .

II)

- 1 Let $B = A - 2I$. For $n \in \mathbb{N}^*$, compute B^n in terms of B .
- 2 Deduce A^n in terms of n , A and I .

III)

- 1 Show that $A^2 - 3A + 2I = 0$.
- 2 Prove by induction that there exist two sequences (a_n) and (b_n) such that, for all integers n ,

$$A^n = a_n A + b_n I$$

Give the recurrence relations satisfied by (a_n) and (b_n) and determine a_n and b_n in terms of n .
Deduce the expression of A^n in terms of n , A , and I .

- 3 Justify that A is invertible and give its inverse.
- 4 Find the same expression of A^n by performing the Euclidean division of X^n by $X^2 - 3X + 2$.

Exercise 2

Compute the powers of the following matrices : $\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$, $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

Exercise 3

Let θ be a real number.

1 We define

$$R_\theta = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

Compute R_θ^k for any natural number k .

2 Consider the sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ defined by :

$$\begin{cases} x_0 = 1 \\ y_0 = 1 \\ \forall n \in \mathbb{N}, & x_{n+1} = \cos(\theta)x_n - \sin(\theta)y_n \\ \forall n \in \mathbb{N}, & y_{n+1} = \sin(\theta)x_n + \cos(\theta)y_n. \end{cases}$$

Using the previous question, find explicit expressions for the sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$.

Exercise 4

Are the following matrices invertible? If so, compute their inverse.

$$\begin{pmatrix} \cos a & 0 & -\sin a \\ 0 & 1 & 0 \\ \sin a & 0 & \cos a \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ -2 & 3 & -1 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \ddots & 1 \\ 1 & 0 & \dots & \dots & 0 \end{pmatrix}.$$

Exercise 5

Determine the rank of the following matrices depending on the parameters :

$$A = \begin{pmatrix} 1-a & 0 & 0 \\ -1 & 2-a & 1 \\ 2 & 0 & 3-a \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ bc & ca & ab \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{pmatrix},$$

$$D = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ 2 & 3 & 4 & \dots & n+1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n & n+1 & n+2 & \dots & 2n-1 \end{pmatrix}, \quad E = \begin{pmatrix} a & 1 & \dots & \dots & 1 \\ 1 & a & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 1 \\ 1 & \dots & \dots & 1 & a \end{pmatrix}$$

Exercise 6

Let $A \in GL_n(\mathbb{K})$, $C \in GL_p(\mathbb{K})$, and $B \in \mathcal{M}_{n,p}(\mathbb{K})$.

Show that the block matrix (partitioned matrix) $\begin{pmatrix} A & B \\ 0_{p,n} & C \end{pmatrix} \in \mathcal{M}_{n+p}(\mathbb{K})$ is invertible, and determine its inverse.

Exercise 7

- 1 Show that any square matrix can be uniquely decomposed as the sum of a symmetric matrix and an antisymmetric matrix.
- 2 Let two symmetric matrices be given. Show that their product is symmetric if and only if the two matrices commute.
- 3 Show that, when it exists, the inverse of an antisymmetric matrix is also antisymmetric.

Exercise 8

Let $A \in GL_n(\mathbb{R})$ such that $A + A^{-1} = I_n$. Determine $A^k + A^{-k}$ for all $k \in \mathbb{N}$.

Exercise 9

Let $A = \begin{pmatrix} 1 & 1 & \lambda \\ \lambda - 1 & -1 & -1 \\ 0 & 2\lambda & 1 \end{pmatrix}$. Determine the values of $\lambda \in \mathbb{R}$ for which $A \in GL_3(\mathbb{R})$. For these values, compute A^{-1} . Do there exist complex values of λ that make A non-invertible in $M_3(\mathbb{C})$?

Exercise 10

Let $\text{tr}(A)$ denote the trace of a square matrix A , which is the sum of its diagonal entries.

- 1 Show that if A and B are two square matrices of the same size, and λ is a real number, the following relations hold :
 - a $\text{tr}(\lambda A) = \lambda \text{tr}(A)$
 - b $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
 - c $\text{tr}(AB) = \text{tr}(BA)$
- 2 Let $A \in \mathcal{M}_n(\mathbb{K})$. Show that the equation $AX - XA = I_n$, with unknown $X \in \mathcal{M}_n(\mathbb{K})$, has no solution.

Exercise 11

Consider the matrix $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

- 1 Compute A^k for $k \in \mathbb{N}^*$.
- 2 Let $P \in \mathbb{R}_n[X]$, with $n \geq 2$. Prove that :

$$P(A) = \begin{pmatrix} P(2) & P'(2) & \frac{1}{2}P''(2) \\ 0 & P(2) & P'(2) \\ 0 & 0 & P(2) \end{pmatrix}.$$

Exercise 12

Let $n \in \mathbb{N}^*$. A matrix $A \in M_n(\mathbb{R})$ is said to be **nilpotent** if there exists $k \in \mathbb{N}$ such that $A^k = 0_n$. In this exercise, we consider a **nilpotent** matrix $A \in M_n(\mathbb{R})$.

- 1 Show that there exists $m \in \mathbb{N}^*$ such that $A^{m-1} \neq 0_n$ and $A^m = 0_n$. This integer m is called the **nilpotency index** of A .
- 2 Show that A is not invertible.
- 3 Show that for any nonzero natural number p , we have :

$$A^p - I_n = (A - I_n) \sum_{k=0}^{p-1} A^k.$$

Deduce that $I_n - A$ is invertible and compute its inverse.

- 4 Deduce the inverse of $M = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ (without using Gaussian elimination).
- 5 Show that $I_n - A^{-1}$ is nilpotent.

Exercise 13 : Proposed by the student Abderrahim Cherfaoui

Let A, B and C be $n \times n$ matrices with complex entries satisfying

$$A^2 = B^2 = C^2 \quad \text{and} \quad B^3 = ABC + 2I.$$

Prove that $A^6 = I$.